

Benchmarking Foundation Models using Satellite Image Time Series for Land Monitoring - MVA 2025/2026

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March 26, 2026

Table 1 Test metrics

	mIoU	F1 Macro	Overall accuracy
test_alpha_earth_ruche_h512_h256_lr0.001	0.5287	0.6526	0.8417
test_tessera_ruche_h512_h256_lr0.001	0.5607	0.7018	0.8257
test_alise_ruche_h512_h256_lr0.001	0.6009	0.7235	0.8651
utae_ruche_train_decoder_128_encoder_128_out_20c	0.7219	-	0.9170

Table 2 Test F1 per class

	test_alpha_earth_ruche_h512_h256_lr0.001	test_tessera_ruche_h512_h256_lr0.001	test_alise_ruche_h512_h256_lr0.001
Meadow	0.9217	0.8859	0.9216
Soft winter wheat	0.8793	0.8501	0.9155
Corn	0.9184	0.8572	0.9306
Winter barley	0.6718	0.7947	0.8354
Winter rapeseed	0.9124	0.8734	0.9361
Spring barley	0.5308	0.6404	0.7050
Sunflower	0.6303	0.7465	0.6896
Grapevine	0.8745	0.8561	0.8803
Beet	0.8953	0.8247	0.9289
Winter triticale	0.3222	0.4492	0.4334
Winter durum wheat	0.7250	0.7168	0.7979
Fruits, vegetables, flowers	0.4885	0.5821	0.5547
Potatoes	0.6144	0.5599	0.6988
Leguminous fodder	0.4339	0.5538	0.4998
Soybeans	0.8129	0.8238	0.8646
Orchard	0.7246	0.6713	0.6528
Mixed cereal	0.1472	0.4740	0.3414
Sorghum	0.2433	0.4725	0.4367

1 Introduction

2 Dataset

We use the dataset PASTIS (Panoptic Agricultural Satellite Time Series) There are 2433 sequences of multi-spectral of shape 10*128*128. Each sequence contains between 38 and 61 images. The satellite THEIA provides observations. The 10 channels correspond to the non-atmospheric spectral bands of the Sentinel-2 satellite, after atmospheric correction and re-sampling at a spatial resolution of 10 meters per pixel. Observations are taken in several regions of France with different climates, (Normandie, Bourgogne, Alsace, Provence). Almost 30 % of pictures are covered by cloud.

Each pixel is associated with a semantic label of 18 crop labels

3 Methods

4 Results

[1] [2]

5 Summary of findings

Table 3 Test metrics with scarce data (30 patches per fold)

	mIoU	F1 Macro	Overall accuracy
test_alpha_earth_ruche_scarce_h512_h256_lr0.001	0.2867	0.3755	0.7267
test_tessera_ruche_scarce_h512_h256_lr0.001	0.4387	0.5765	0.7621
test_alise_ruche_scarce_h512_h256_lr0.001	0.4482	0.5729	0.7898
test_utae_ruche_scarce_s2_10ch	0.3869	0.5115	0.7043

References

- [1] Zhengpeng Feng et al. *TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis*. 2025. arXiv: [2506.20380](https://arxiv.org/abs/2506.20380) [cs.LG]. URL: <https://arxiv.org/abs/2506.20380>.
- [2] Christopher F. Brown et al. *AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data*. 2025. arXiv: [2507.22291](https://arxiv.org/abs/2507.22291) [cs.CV]. URL: <https://arxiv.org/abs/2507.22291>.

Appendix A Method

Appendix B Data